

TW17: Final Project Presentations and Review

Five challenges | 20 marks each | 100 marks total

Student name	
Student ID	
Class / group	
Submission date	

Weekly purpose

Complete and present a reproducible end-to-end project with a quality gate, evidence-led visual story, validated method, ethical review, and final handover.

SUBMISSION RULES

- Complete work in the corresponding Jupyter notebook.
- Write your own code and explanations; no solutions are supplied.
- Keep supplied data unchanged and work on copies.
- Restart the kernel and run all cells before submission.
- Submit the notebook with outputs visible and assistance acknowledged.

MARKING RUBRIC FOR EACH CHALLENGE

Assessment area	Marks
Correct method and working code	8
Validation and quality checks	4
Explanation and interpretation	4
Reproducibility and presentation	4
Total per challenge	20

Challenge 1: Approve the project proposal and team plan

Scenario

The group must agree scope before analysis begins.

INSTRUCTIONS

1. Choose one supplied dataset and a specific decision-led question.
2. Define audience, unit, outcome, success criteria, excluded uses, and deliverables.
3. Assign lead, data steward, analyst, modeller, reviewer, and presentation responsibilities.
4. Create a seven-day work plan with review checkpoints.

REQUIRED SUBMISSION

- A signed project contract.
- A responsibility matrix and schedule.
- At least four limitations or risks identified before modelling.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 2: Build the quality gate and cleaning log

Scenario

The project cannot proceed until its data assumptions are verified.

INSTRUCTIONS

1. Record source, format, schema, row count, unit of analysis, and key.
2. Check required columns, key uniqueness, missingness, duplicates, categories, dates, and ranges.
3. Separate critical failures from warnings.
4. Implement cleaning on a copy and record issue, evidence, action, reason, and impact.
5. Write final data-contract assertions.

REQUIRED SUBMISSION

- A quality-gate report.
- A reproducible cleaning function and decision log.
- Passing final assertions with raw data preserved.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 3: Create the exploratory evidence story

Scenario

The presentation needs only the evidence required to answer the project question.

INSTRUCTIONS

1. Produce one overall summary and at least one grouped comparison.
2. Use robust statistics where distributions are skewed.
3. Create two complementary, accessible charts.
4. Run one sensitivity analysis.
5. Write one finding, one alternative explanation, and one limitation.

REQUIRED SUBMISSION

- Two checked statistical tables.
- Two presentation-ready charts with units and scope.
- A finding supported by output and a sensitivity result.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 4: Validate the analytical method and ethics

Scenario

The group must justify either supervised learning, clustering, time-series analysis, or a statistical test.

INSTRUCTIONS

1. State why the method matches the question and variable types.
2. Define a baseline or comparison and an appropriate validation design.
3. Report relevant metrics with variability or uncertainty.
4. Inspect important error cases or unstable assignments.
5. Create an ethics and operational risk register with mitigations.

REQUIRED SUBMISSION

- A reproducible validated method.
- Baseline/comparison and evidence table.
- A risk register covering privacy, bias, human oversight, and monitoring.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 5: Present, peer-review, and hand over

Scenario

The final analysis must be understandable and reproducible by another team.

INSTRUCTIONS

1. Create a seven-slide, seven-minute storyboard from decision to next test.
2. Use the 100-point rubric for peer scoring and record actionable evidence.
3. Run the notebook from a clean kernel on another machine or environment.
4. Audit every major claim against output and study design.
5. Create a final handover record with files, environment, data version, seeds, approved use, excluded use, limitation, and monitoring need.

REQUIRED SUBMISSION

- A timed storyboard and completed peer rubric.
- A reproducibility and claim audit with resolved failures.
- A final handover record and 200-word executive conclusion.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Final submission checklist

- ☐ All five challenges are complete.
- ☐ The notebook runs from top to bottom without errors.
- ☐ Every output is clearly labelled.
- ☐ Important results include written explanations.
- ☐ Validation checks are included.
- ☐ Supplied raw data has not been overwritten.
- ☐ Sources and assistance are acknowledged.

Student declaration

I confirm that this submission is my work, that I have acknowledged sources and assistance, and that the displayed outputs were produced by the submitted code.

Signature: _____ Date: _____